

HITESH KUMAR PUNNA

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SUMMARY

Quant researcher with experience in systematic **options strategies** and ML-driven signal tools. Pursuing a Master's in Financial Engineering and seeking roles blending **machine learning** and **quantitative finance**.

EDUCATION

GEORGIA INSTITUTE OF TECHNOLOGY, Scheller College of Business Atlanta, GA
Master of Science in Quantitative and Computational Finance Aug 2025 – May 2027

- GPA: N/A (joining Fall 2025)
- Key Coursework: C++, Stochastic Calculus, Derivatives, Statistical Learning, Fixed Income, Numerical Methods

INDIAN INSTITUTE OF TECHNOLOGY BOMBAY Mumbai, India
Bachelor of Technology Jul 2019 – May 2023

- GPA: 8.86/10
- Major in Computer Science and Engineering with Honors and Minor in Applied Statistics

EXPERIENCE

GRAVITON RESEARCH CAPITAL LLP Delhi, India
Quant Researcher and Analyst Jun 2023 – Jul 2025

- Built and deployed a high-performing **stock options strategy** on the NSE, managing the full pipeline from signal research to live execution and transitioning it into production with consistent returns.
- Scaled an existing commodity options strategy on **MCX**, expanding from 2 to 6 instruments including **GOLD, SILVER**, and Mini contracts, resulting in a **2× increase in strategy profitability**.
- Developed **XGBoost classifiers** trained on 50+ features (liquidity, Greeks, volatility) to denoise signals, identify market regimes, and optimize entries resulting in a **10–20% gain in net PnL**.
- Automated hyperparameter tuning using **Genetic Algorithms**, reducing model training time from **120 minutes to under 15 minutes** and accelerating the research-to-deployment cycle.
- Modeled complex derivatives such as **Callable Bull/Bear Contracts (CBBs)** using multi-factor **linear regression**, improving pricing accuracy and risk control in low-liquidity environments.

GRAVITON RESEARCH CAPITAL LLP Delhi, India
Quant Research Intern May 2022 – Jul 2022

- Designed microstructure based market indicators using statistical filters to capture short-term trend strength and intraday regime shifts, aiding signal generation in high-frequency environments.
- Trained and deployed ML models to extend traditional linear regression frameworks using existing alphas and few market indicators; boosted NSE Index Options strategy PnL by **10%** in backtesting.
- Enhanced alpha performance for both stock and index options books by tuning weighting functions and suppressing poor regimes, improving returns from **6% to 8%** and **8% to 10%**, respectively.

EXO FLY (Tech Team IIT Bombay) Mumbai, India
Student Team Lead Mar 2022 - Apr 2023

- Led a **6-member team** and secured funding to design a safe and compact manned **eVTOL aerial vehicle**
- Designed a controller on **Simulink**, performing **sensor fusion** for localization with an **Extended Kalman Filter**

PROJECTS

Optimization of Deep Neural Networks via Layer Pruning Spring 2023
Guide: Prof. Rushikesh K.Joshi BTP Project

- Employed **layer and node pruning**, and training **student-teacher networks**, to optimize the size of neural networks.
- Achieved a significant one-thirds reduction in model size while ensuring accuracy remained within a 5% margin.

MDP Optimization with Reinforcement Learning Autumn 2021
Guide: Prof. Shivaram Kalyanakrishnan Course Project

- Explored sampling algorithms like **KL-UCB** and **Thompson Sampling** for stochastic multi-armed bandits
- Programmed **SARSA** with linear approximation for the mountain car game using **OpenAI Gym** in Python

SKILLS

Programming Languages C++, Python, Bash, PyTorch, TensorFlow, scikit-learn, OpenCV, MATLAB
English, Hindi